ClipSAT : Cambridge & Edexcel Additional Mathematics 0606/4PM1

Unit 7.01: Differentiation & Rates of Change

Name: _____

Q1. Differentiate with respect to x : y = 3x^{3} - 5x + 7 .

[ ] A: frac{dy}{dx} = 3x^{2} - 5
[ ] B: frac{dy}{dx} = 9x^{2}
[ ] C: frac{dy}{dx} = 9x^{3} - 5
[ ] D: frac{dy}{dx} = 9x^{2} - 5

Q2. Differentiate with respect to x : y = 6x^{4} - 5x + 7 .

[ ] A: frac{dy}{dx} = 24x^{3} - 5
[ ] B: frac{dy}{dx} = 24x^{4} - 5
[ ] C: frac{dy}{dx} = 6x^{3} - 5
[ ] D: frac{dy}{dx} = 24x^{3}

Q3. Differentiate with respect to x : y = 9x^{5} - 5x + 7 .

[ ] A: frac{dy}{dx} = 45x^{4} - 5
[ ] B: frac{dy}{dx} = 45x^{5} - 5
[ ] C: frac{dy}{dx} = 9x^{4} - 5
[ ] D: frac{dy}{dx} = 45x^{4}

Q4. Differentiate with respect to x : y = 12x^{6} - 5x + 7 .

[ ] A: frac{dy}{dx} = 72x^{5}
[ ] B: frac{dy}{dx} = 12x^{5} - 5
[ ] C: frac{dy}{dx} = 72x^{6} - 5
[ ] D: frac{dy}{dx} = 72x^{5} - 5

Q5. Differentiate with respect to x : y = 15x^{7} - 5x + 7 .

[ ] A: frac{dy}{dx} = 105x^{6}
[ ] B: frac{dy}{dx} = 15x^{6} - 5
[ ] C: frac{dy}{dx} = 105x^{7} - 5
[ ] D: frac{dy}{dx} = 105x^{6} - 5

Q6. Differentiate with respect to x : y = 18x^{8} - 5x + 7 .

[ ] A: frac{dy}{dx} = 144x^{7}
[ ] B: frac{dy}{dx} = 144x^{7} - 5
[ ] C: frac{dy}{dx} = 144x^{8} - 5
[ ] D: frac{dy}{dx} = 18x^{7} - 5

Q7. Find the equation of the tangent line to the curve y = 2x + 1^3 at the point where x = 0 .

[ ] A: y = 2x + 1
[ ] B: y = 6x - 1
[ ] C: y = 6x + 1
[ ] D: y = 6x + 3

Q8. Find the equation of the tangent line to the curve y = 3x + 1^3 at the point where x = 0 .

[ ] A: y = 9x + 3
[ ] B: y = 9x - 1
[ ] C: y = 3x + 1
[ ] D: y = 9x + 1